Conceptual Design Report for Part 4B

Power Management and DC–DC Distribution Network

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1 Introduction and Objectives

This report presents a *conceptual* power management and distribution architecture for Part 4B. The goal is to design a DC–DC power distribution network that delivers a total of *approximately* 750 W from a nominal 28 V DC input (military aircraft / vehicle bus) to multiple regulated rails supporting an FPGA-centric system, digital logic, analog front-end, RF stages, and power amplifiers.

The focus here is on **design choices and qualitative justification**, rather than on detailed numerical design or sizing. No component values, thermal figures, or efficiency numbers are finalized; instead, the report explains which topologies are selected, how rails are grouped, what control and protection features are required, and what kind of components will appear in a future detailed design.

Key high-level objectives:

- Achieve high overall efficiency to minimize waste heat in an enclosed, harsh environment.
- Meet tight ripple and noise requirements for sensitive FPGA and RF/analog circuits.
- Implement safe sequencing and protection to avoid damaging high-value components.
- Use realistic, vendor-supported topologies and ICs that are appropriate for military conditions (voltage variation, temperature, transients).

2 System Requirements (Qualitative)

The system is powered from a 28 V nominal input, which is assumed to be subject to:

- A defined operating voltage range (e.g., a lower and upper bound around 28 V).
- Transient events such as dips, surges, and load dump consistent with aircraft/vehicle power bus standards.
- Environmental extremes in temperature and vibration typical of military platforms.

The power distribution network must generate multiple rails, including:

- A low-voltage, very high-current rail for the FPGA core (on the order of 1 V with very high current).
- A medium-current rail for FPGA auxiliary functions (around 1.8 V).
- A digital logic rail (around 3.3 V).
- One or more analog rails (e.g., 5 V and 12 V) with tighter noise constraints.

- A higher-voltage rail (close to 28 V) for RF/power amplifiers, which may be derived with minimal conversion.

Qualitatively, the FPGA core rail is the most challenging because:

- It operates at the lowest voltage but highest current, making copper losses and voltage drops critical.
- It is highly dynamic, with fast load transients dependent on FPGA internal activity.
- It has tight tolerance and ripple requirements to avoid timing violations and functional instability.

Analog and RF rails are sensitive to noise and ripple because:

- Switching noise can modulate sensitive analog signals.
- Phase noise and amplitude modulation can directly degrade RF performance.

3 Topology Selection and Architecture

3.1 Overall Strategy

Given that all required output voltages are lower than the 28 V input, the primary conversion topology for each rail is a **step-down (buck) converter**. The architecture is structured as follows:

- A **multi-phase synchronous buck** converter for the low-voltage, high-current FPGA core rail.
- **Single-phase synchronous buck** converters for the medium-current digital and auxiliary rails.
- **Switching pre-regulation plus linear post-regulation** for low-noise analog rails (e.g., 5 V and possibly 12 V).
- A largely **direct 28 V distribution path** for power amplifiers, with appropriate filtering and protection rather than conversion to a lower voltage and back up.

This partitioning is chosen to:

- Concentrate complexity where it has the greatest impact (FPGA core and key analog rails).
- Minimize conversion stages for high-power paths to maintain efficiency.
- Keep the number of unique controllers manageable, simplifying design and validation.

3.2 FPGA Core Rail: Multi-Phase Synchronous Buck

Choice: A multi-phase synchronous buck converter is selected for the FPGA core rail.
Justification:

- **Current sharing:** Splitting the total core current across multiple phases keeps per-phase current within realistic limits for available MOSFETs and inductors.
- **Reduced ripple:** Interleaving phases cancels a significant portion of the output ripple, easing output capacitor requirements and improving transient behavior.

- **Improved thermal distribution:** Losses are spread across multiple MOSFETs and magnetics, leading to lower local hot spots and simpler heatsinking.
- **Better transient response:** Multi-phase architectures can respond faster to load steps because each phase contributes to current ramp-up.
- **Synchronous rectification:** High duty cycle and current levels make diode conduction losses unacceptable; synchronous MOSFETs drastically reduce losses.

A dedicated multi-phase controller from a major vendor (e.g., Texas Instruments, Analog Devices, Infineon, etc.) would be chosen to:

- Provide phase interleaving and current balancing.
- Integrate protection features (over-current, over-voltage, under-voltage lockout, fault reporting).
- Offer a stable compensation framework tailored for multi-phase operation.

3.3 FPGA Auxiliary and Digital Rails: Single-Phase Synchronous Buck

Choice: Single-phase synchronous buck converters are selected for the auxiliary FPGA rail and the 3.3 V digital rail.

Justification:

- These rails draw significant but moderate current, making single-phase solutions more than adequate.
- Synchronous rectification is still advantageous to maintain efficiency, especially under continuous operation.
- Integrated controller + FET solutions or controller ICs with external MOSFETs offer a good trade-off between flexibility, efficiency, and design complexity.
- Using similar families of controllers across rails simplifies development, layout reuse, and firmware interaction (if digital control is used).

3.4 Analog and RF Rails: Switching Plus Linear Post-Regulation

Choice: A two-stage approach:

1. A switching pre-regulator (buck) to an intermediate voltage (e.g., 5 V or 12 V).
2. Low-dropout (LDO) or ultra-low-noise linear regulators downstream for critical analog loads.

Justification:

- Pure linear regulation from 28 V to low voltages at meaningful currents would be excessively inefficient and generate too much heat.
- By first stepping down using a high-efficiency switcher, the voltage drop and power dissipated in the linear stage are minimized.
- LDOs provide excellent power-supply rejection and low output noise, essential for ADCs, PLLs, and sensitive RF stages.
- This hybrid approach balances efficiency with noise performance.

For RF power amplifiers requiring a 28 V-ish supply, a primarily *direct* path is preferred:

- Minimize unnecessary conversion stages to preserve efficiency.
- Use LC filters and potentially small series elements to filter bus ripple and switching noise originating elsewhere in the system.
- Ensure that the distribution network can handle the high current and manage voltage drops along traces and planes.

4 Sequencing and Protection Strategy

4.1 Power-On Sequencing

Choice: Use a dedicated power sequencing / supervisor IC or a microcontroller-based sequencing scheme.

Justification:

- FPGAs typically require that I/O-related rails (e.g., 3.3 V, 1.8 V) are at valid levels before the core rail is applied, to avoid latch-up and unintended current paths through protection diodes.
- A deterministic power-up order reduces risk of damage and undefined behavior at start-up.
- A dedicated sequencing IC can monitor all rails, enforce delays between them, and shut down rails in a defined order if a fault occurs.
- Using an IC with programmable thresholds and timing parameters allows easy adjustment late in the design without hardware changes.

The qualitatively defined sequence is:

1. Input 28 V rail becomes valid and is checked for range.
2. Digital / I/O rails (e.g., 3.3 V) ramp up and stabilize.
3. Auxiliary FPGA rails (e.g., 1.8 V) follow after a small delay.
4. Core rail (e.g., 1.0 V) is enabled last, after I/O rails are confirmed within tolerance.

4.2 Protection Features

Over-Current Protection (OCP):

- Each converter includes cycle-by-cycle or averaged current limiting.
- Foldback or hiccup behavior is preferred over simple "hard limit" to reduce stress during faults.

Over-Voltage Protection (OVP):

- Supervisory circuits monitor each rail and shut down the corresponding converter if the voltage exceeds a defined threshold.
- On especially sensitive rails (e.g., FPGA core), independent "crowbar" or fast disconnect strategies may be used for additional safety.

Under-Voltage Lockout (UVLO):

- Ensures that converters only start when the input bus is within a valid range.
- Prevents partial turn-on that could cause excessive currents or unpredictable device behavior.

Thermal Protection:

- Many modern controllers include internal thermal shutdown logic.
- External NTC thermistors or temperature sensors near critical devices (MOSFETs, inductors, RF amplifiers) can feed into the sequencer or a supervisory MCU, enabling graceful power derating or shutdown.

5 Rough Bill of Materials (Conceptual)

The following table lists typical *types* of components that would appear in a realized design. No specific values or part numbers are finalized at this stage.

Category	Description (Conceptual)
Multi-Phase Controller IC	Dedicated multi-phase buck controller for FPGA core rail (supports several phases, current balancing, protection).
Phase MOSFETs	High-current, low $R_{DS(on)}$ N-channel MOSFETs for high-side and low-side of each phase.
Phase Inductors	Power inductors rated for the per-phase current of the core rail, with low DC resistance and suitable saturation characteristics.
Core Rail Output Capacitors	Bank of low-ESR ceramic capacitors, possibly supplemented by polymer capacitors, placed near FPGA power pins.
Single-Phase Buck Controllers	Controller (or integrated regulator) ICs for auxiliary and digital rails (e.g., 1.8 V and 3.3 V).
Analog Pre-Regulator	Switching regulator IC for analog rails (e.g., 5 V or 12 V), selected for low noise and good transient performance.
Low-Noise LDOs	Linear regulators for final analog and RF rails where very low noise and ripple are essential.
Input/Output Filter Components	Inductors, ferrite beads, and capacitors for input and output EMI filtering on critical rails.
Sequencing / Supervisor IC	Multi-channel power sequencer or supervisor IC, capable of enforcing order, timing, and fault response across rails.
Temperature Sensors / NTCs	Sensors or thermistors placed near hot components for thermal monitoring and protection.
Connectors (MIL Style)	MIL-qualified power connector for 28 V input, plus board-to-board or harness connectors as needed.
PCB and Mechanical Hardware	Multi-layer PCB with thicker copper on power layers, heatsinks or thermal pads attached to MOSFETs and possibly inductors.

Table 1: Conceptual (Rough) Bill of Materials – No values or part numbers finalized

6 Conclusion

This conceptual report outlines a feasible architecture for the Part 4B power management and distribution network Due to time constraints I failed to complete part B